

AP Computer Science Principles

Week 2a: What Is Computer Science?

Today's Objectives:

- Students will be able to define computer science and its scope.
- Students will be able to explain the positive impacts of computer science in various domains.

Welcome to Computer Science

What does a computer scientist do?



“Official” definition of computer science:

Computer science is the study of algorithms and computational systems, including their design, development, and applications.



But what does this mean? First, let’s look at some common misconceptions when it comes to the computer science field:

Misconception 1: Computer science is the study of computers

- Computer Science is no more about computers than chemistry is about beakers and test tubes.
- Science is not about the tools that are used.
- It is about how we use them and what we find out when we do.



Misconception 2: Computer science is the study of how to write computer programs



- Programming is extremely important to the discipline but programming is a tool.
- The construction of a high-quality program is important but what the program does and the services it provides are just as important.

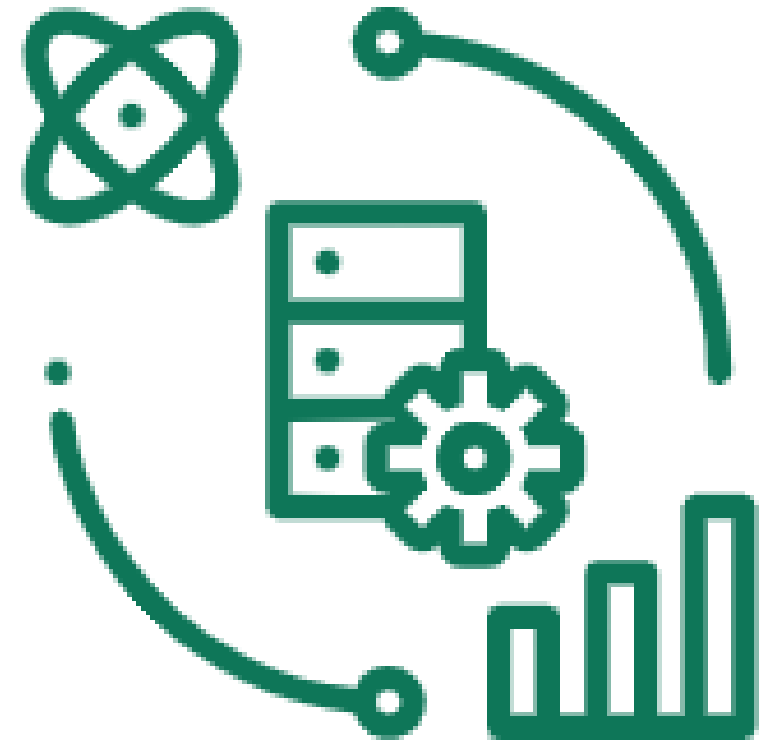
Misconception 3: Computer science is the application of computer software.

- A wide range of people use computer software but it is the computer scientist who is responsible for specifying, designing, building, and testing software.



Computer Science Shapes Our World

- Computational devices and systems have infiltrated our world, for better or worse.
- Involving computers in our everyday lives has improved many things, and in this course, we will discuss the good and the bad.
- Let's look at some positive impacts computer science has had on four pillars of our society: healthcare, entertainment, transportation, and communication.



<i>Healthcare</i>	<i>Entertainment</i>
• Computational modeling aids in drug discovery. • Machine learning algorithms help diagnose diseases more accurately. • Electronic health records have led to better care coordination.	• Computer graphics and animation are at the core of most entertainment experiences. • Social media platforms facilitate global connectivity. • Streaming platforms use machine learning to provide personalized content.

<i>Transportation</i>	<i>Communication</i>
• Mapping apps use algorithms to provide accurate route information. • Traffic management systems help optimize traffic flow. • Autonomous vehicles are being developed to enable safer and more efficient transportation.	• The Internet revolutionized global communication. • Encryption and cybersecurity technologies have allowed for secure e-commerce experiences. • Natural language processing has allowed for global communication to be more accessible.

Computer Science Fields

- Computer Science is not a single field, but is an umbrella subject area that many careers fall under.
- Here are some (not all) examples of possible careers computer science can lead you to:

Software Engineer:

- Software engineers design, develop, and maintain software applications.
- They work on everything from mobile apps to operating systems.

Data Scientist:

- Data scientists analyze and interpret large sets of data to extract insights and inform decision-making.
- They apply statistical analysis, machine learning, and data visualization techniques to solve complex problems.

Artificial Intelligence (AI) Engineer:

- AI engineers build and deploy intelligent systems and algorithms.
- They work on tasks like natural language processing, computer vision, and machine learning to develop applications such as virtual assistants, recommendation systems, and autonomous vehicles.

Cybersecurity Analyst:

- Cybersecurity analysts protect computer systems and networks from potential threats and vulnerabilities.
- They develop security protocols, monitor and respond to incidents, and conduct risk assessments to ensure the integrity and confidentiality of data.

Web Developer:

- Web developers design and build websites using HTML, CSS, and JavaScript.
- They create visually appealing and user-friendly interfaces and may specialize in front-end or back-end development.

Game Developer:

- Game developers design and create video games.
- They develop game mechanics, implement graphics and sound, and collaborate with artists and designers to bring interactive experiences to life.

Important Computer Science Terms

We have introduced some very important vocab terms with this lesson. Let's review their definitions:

- **Computer Software:** collection of programs, data, and instructions that enable a computer system to perform specific tasks or functions
- **Computer Hardware:** physical components of a computer system that can be touched and manipulated.



Important Computer Science Terms

- **Computational System**: combination of hardware, software, and data that work together to perform computational tasks
- **Computational Device**: any device capable of performing computations or processing information
- **Machine Learning**: a way of training computer models on large datasets so they can recognize patterns, make inferences, and improve with experience

